

Name _____

Q1) What is the output for this code segment?

```
int x=5,y=2, temp=0;
temp=x; x=y; y=temp;
cout<<x<<y;
```

Output

- a) 2 b) 5 c) 25 d) other

Q2) what is the output?

```
void swap(int *p, int *q){
    int t=0; t=*p; *p=*q; *q=t;
}
int main(){
    int a=1,b=2;    cout<<a<<b;
    swap(&a,&b);    cout<<a<<b;
}
```

Output

- a) 1221 b) 1212 c) 2121 d) other

Q3) what is the output ?

```
template <class T>
void swap(T *p, T *q){
    T t=0; t=*p; *p=*q; *q=t;
}
int main(){
    int a=3,b=8;    cout<<a<<b;
    swap(&a,&b);    cout<<a<<b;
}
```

Output

- a) 3883 b) 3838 c) 8383 d) other

Q4) What is the output?

```
void swap(char &m, char &n){
    char t=0; t=m; m=n; n=t;
    //cout<<m<<n<<endl;
}

int main(){
    char a='a', b='b'; cout<<a<<b;
    swap(a,b); cout<<a<<b;
}
```

Output

a)ab b)ba c)abba d) other

Q5) what is the output for this code?

```
template <class T>
void swap(T *m, T *n){
    T *t=new int(0); *t=*m; *m=*n; *n=*t;
}

int main(){
    int *a=new int(1); int *b=new int(2);
    cout<<*a<<*b;
    swap(a,b); cout<<*a<<*b;
}
```

Output

a)1221 b)2211 c)2121 d) other

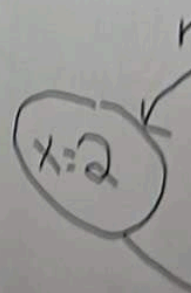
Q6) what is the output?

```
class A{
public:
    int x; A *next;
};

int main()
{
    A *h=new A();
    h->x=2;
    h->next=h;
    cout<<h->x;
    cout<<h->next->next->x;
    return 0;
}
```

Output

a) 2 b)22 c)02 d) other



Q7) what is the output of this list

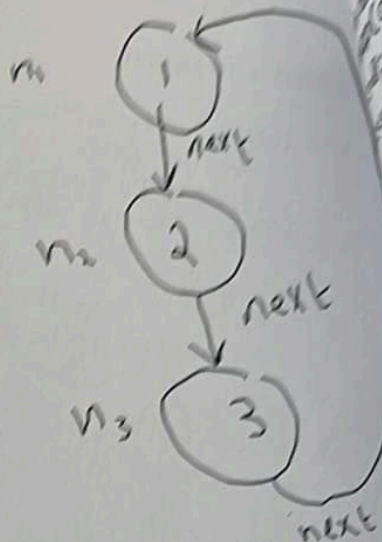
```
class A{
public:
    int g=0;
    A *next;
};

int main()
{
    A *n1=new A(); n1->g=1;
    A *n2=new A(); n2->g=2;
    A *n3=new A(); n3->g=3;
    n1->next=n1; n2->next=n3; n3->next=n1;
    cout<<n3->next->next->next->next->next->g;

    return 0;
}
```

Output

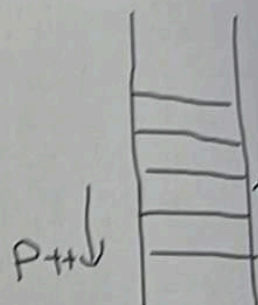
a)1 b)1111 c)123 d)other



Q8) what is the output ?

```
int a[10]={0}, *p=a, counter=0, size=10;
void push(int a){
    if(counter<size){ *p=a; p++; }
    else {cout<<"stack is full";}
}
void pull(){
    if(counter<size){ p--; cout<<*p;}
    else{ cout<<"Stack is empty"<<endl;}
}

int main()
{
    push(9);push(4);push(7); pull();pull();pull();
    return 0;
}
```



Output

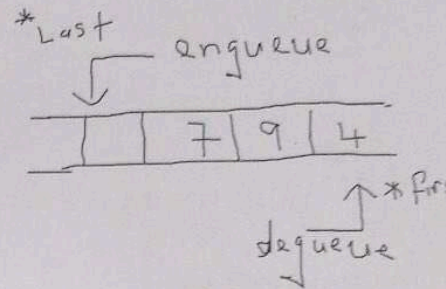
a)777 b)947 c)749 d)other

Q9) what is the output?

```
int a[10]={0},*first=a,*last=a, counter=0, size=10;
void enqueue(int end){
    if(counter<size){ *last=end; last++; counter++; }
    else {cout<<"queue is full";}
}
void dequeue(){
    if(*first !=0){ cout<<*first; first++; counter--;}
    else { cout<<"\n queue is empty"<<endl;}
}
void number(){
    cout<<"number="<<counter<<endl;
}

int main()
{
    enqueue(4); enqueue(9); enqueue(7);
    number();
    dequeue(); dequeue(); dequeue(); dequeue();
    number();

    return 0;
}
```



Write the output

There are some syntax errors in the code, but assuming they are corrected, the output ~~would be~~ would be

number = 3;

4

9

7

queue is empty

number = 0;